

Multi-Agent Collaboration Patterns

Data source: [Du et al., 2023] [Wu et al., 2023] [Hong et al., 2024]

Debate Pattern

Agent A: Argument 1
□
Agent B: Rebuttal
□
Agent A: Revision
□ Converges after rounds
Final Answer

- Improve Factual Accuracy
- Reduce Hallucination
- High multi-round cost
- +15-20% Accuracy

Best for: Reasoning Tasks

Conversation Pattern

PM: Requirements
□
Designer: Prototype
□
Engineer: Challenge
□ Dynamic Synergy
Final Output

- Simulates real team collaboration
- Multi-angle review
- May drift off-topic
- AutoGen Framework

Best for: Complex Tasks

Pipeline Pattern

Step 1: Research
□
Step 2: Writing
□
Step 3: Review
□ Fixed Workflow
Final Output

- Clear, controllable workflow
- Clear roles
- Limited flexibility
- CrewAI Framework

Best for: Standardized Workflows